

FROM SEASONAL STABILITY TO EVENT-DRIVEN DYNAMICS: A SPECTRAL ANALYSIS OF GROUNDWATER RESPONSE TO WINTER CONDITIONS NEAR JFK AIRPORT, NEW YORK



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Abstract:

Long Island depends on a sole-source aquifer that is increasingly vulnerable to road-salt contamination. This study examines groundwater behavior near John F. Kennedy International Airport (JFK), where urbanization and winter maintenance can introduce chloride-rich runoff. Winter conditions were classified using temperature as a proxy for road-salt application, based on the frequency of near-freezing events (-4°C to 1°C). Groundwater-level data (2000–2024) from USGS wells were analyzed using spectral and wavelet methods to evaluate variability across timescales. Results show that winters with more frequent near-freezing conditions exhibit reduced seasonal (low-frequency) variability and increased short-term fluctuations, indicating a shift toward a more dynamic, less buffered aquifer response. These findings suggest that winter environmental conditions can alter groundwater behavior and should be considered in urban groundwater management.

Background Information:

- Long Island relies on a sole-source aquifer for drinking water, making groundwater highly vulnerable to contamination (United States Environmental Protection Agency, 2025; New York State Department of Transportation, 2018).
- Urbanization near JFK Airport, especially impervious surfaces, stormwater systems, and road salt use, significantly alters groundwater recharge and chemistry (Schlesinger & Bernhardt, 2019; Rodbarry et al., 2023).
- Freeze–thaw cycles and snowmelt drive pulses of salty runoff into aquifers, and because chloride is conservative, it persists and elevates background salinity year-round (Stroud Water Research Center, 2026; Schlesinger & Bernhardt, 2019).
- Long Island’s shallow, permeable aquifer allows rapid infiltration of contaminants, impacting drinking water, ecosystems, infrastructure, and metal mobilization (New York State Department of Transportation, 2018; Rodbarry et al., 2023).
- This study uses time-series and spectral analysis to evaluate how winter conditions influence groundwater variability near JFK Airport (Parag et al., 2025).

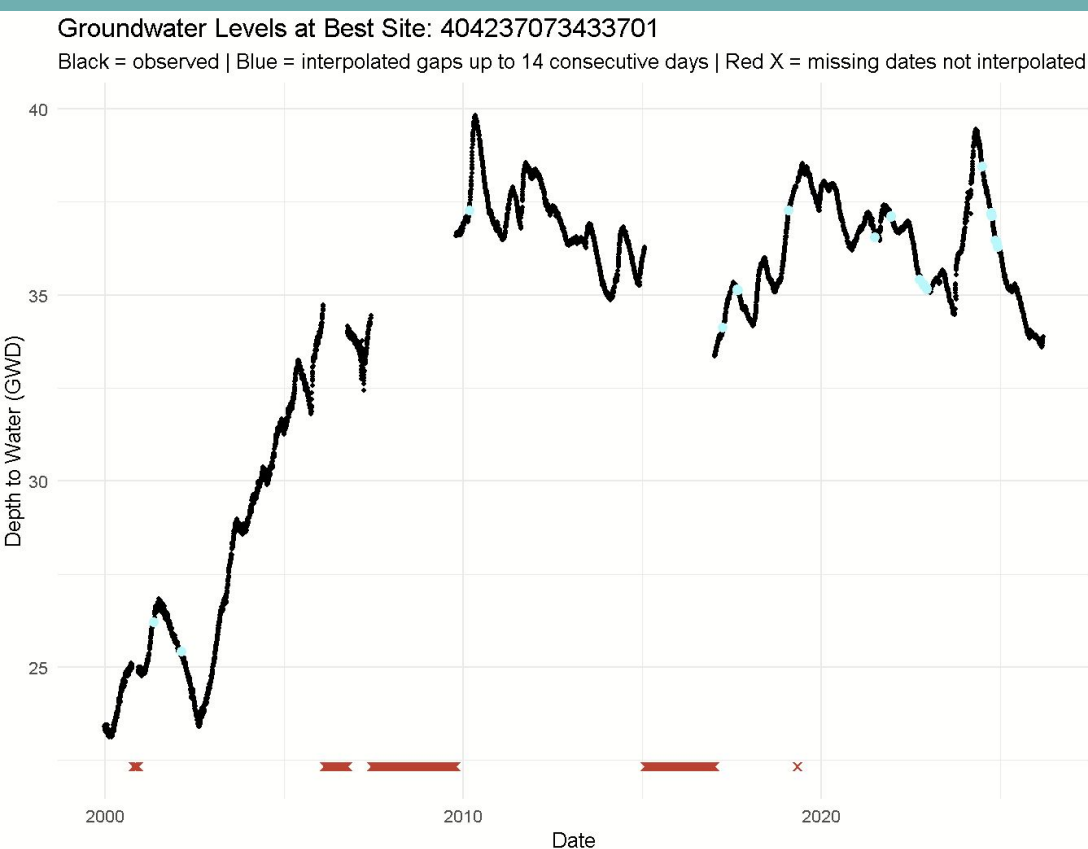


Figure 1: Groundwater level time series (2000–2024) from the well located closest to the highest chloride concentration near JFK Airport, with data quality control applied. Observed data are shown in black, short gaps (≤ 14 consecutive days) filled using linear interpolation in blue, and longer gaps left unfilled (red markers). This approach preserves short-term variability while avoiding artificial smoothing that could bias spectral and wavelet analyses.

Method/Materials:

- USGS NWIS + R (dataRetrieval):**
 - Pulled groundwater level & chloride data (2000–2024) (De Cicco et al., 2023)
- Study Area (JFK, 10 km radius):**
 - Selected wells near JFK using spatial filtering (EPSG: 2263)
- Data Cleaning (R + imputeTS):**
 - Removed duplicates
 - Filled small gaps (≤ 14 days) with linear interpolation
 - Left large gaps unfilled
- Temperature + Precipitation Data (NOAA, GHCNr):**
 - Used temperature to classify winters (freeze–thaw proxy)
 - Used precipitation for context (National Centers for Environmental Information, n.d.)
- Winter Classification (DJF):**
 - Min-salt: no near-freezing days
 - Max-salt: ≥ 1 near-freezing day (-4°C to 1°C) (National Weather Service, n.d.; Hernandez Gonzalez et al., 2023)
- Analysis:**
 - KZ filter (precipitation)
 - Spectral analysis (periodograms, multitaper)
 - Wavelet analysis (time-frequency)
 - Welch t-test for significance (Parag et al., 2025)

Results:

Minimum-salt winters:

- Strong low-frequency (seasonal) signals
- Continuous wavelet bands $\rightarrow$ stable, buffered aquifer response

Maximum-salt winters:

- Reduced seasonal signal + flatter spectra
- Increased short-term variability and fragmented wavelet patterns $\rightarrow$ More irregular, event-driven groundwater behavior

Spectral comparisons:

- Greater variability under maximum-salt conditions, especially at shorter timescales
- Statistically significant difference (Welch t-test, $p < 0.001$)

Spectral power ratios:

- Amplification across all frequencies (strongest at high frequencies) $\rightarrow$ Indicates overall increase in system variability

Precipitation context:

- No long-term trend despite short-term fluctuations $\rightarrow$ Changes reflect aquifer response, not recharge input

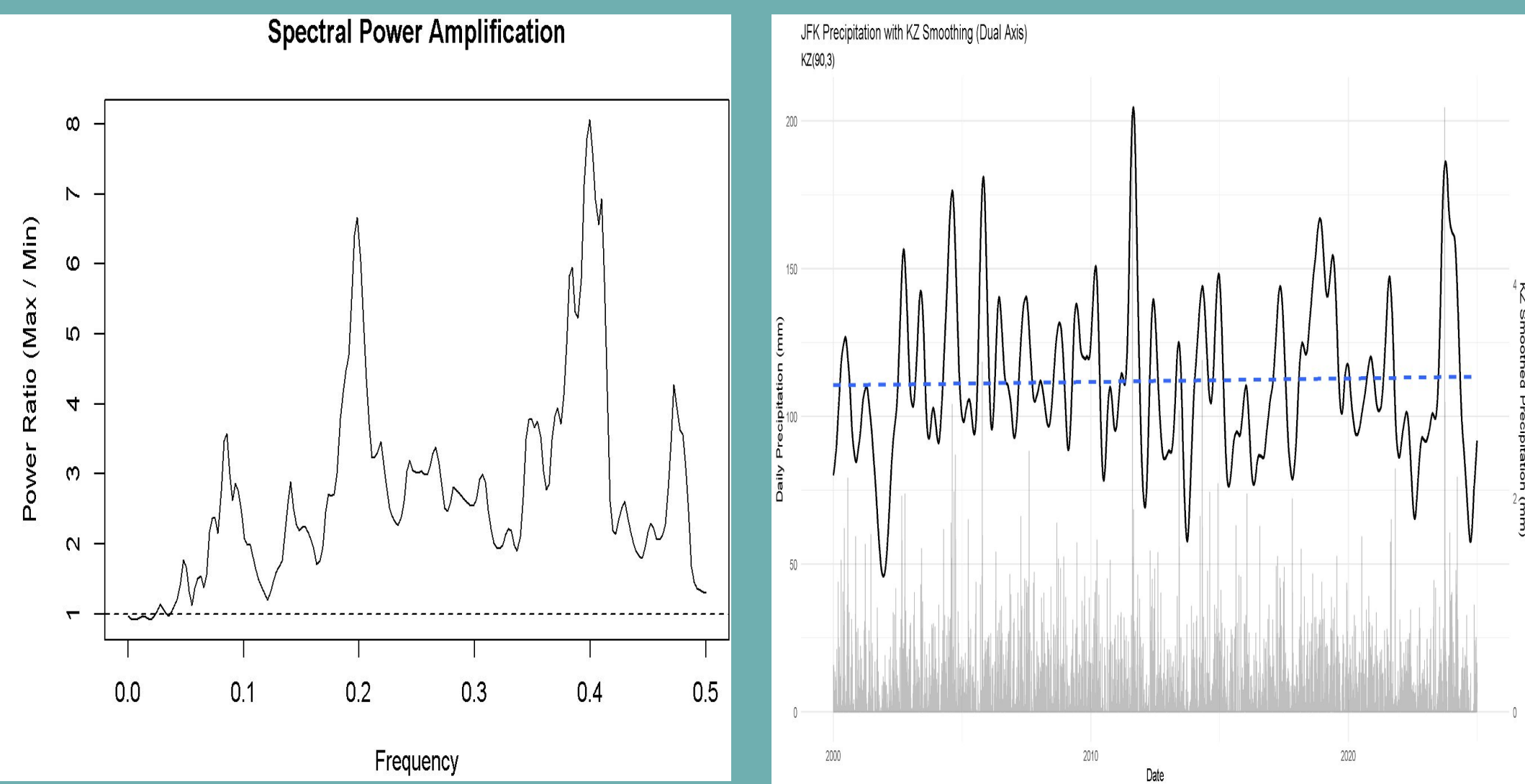


Figure 8: Spectral power ratio (maximum-salt divided by minimum-salt) across frequencies, showing amplification of groundwater variability under maximum-salt conditions, particularly at higher frequencies.

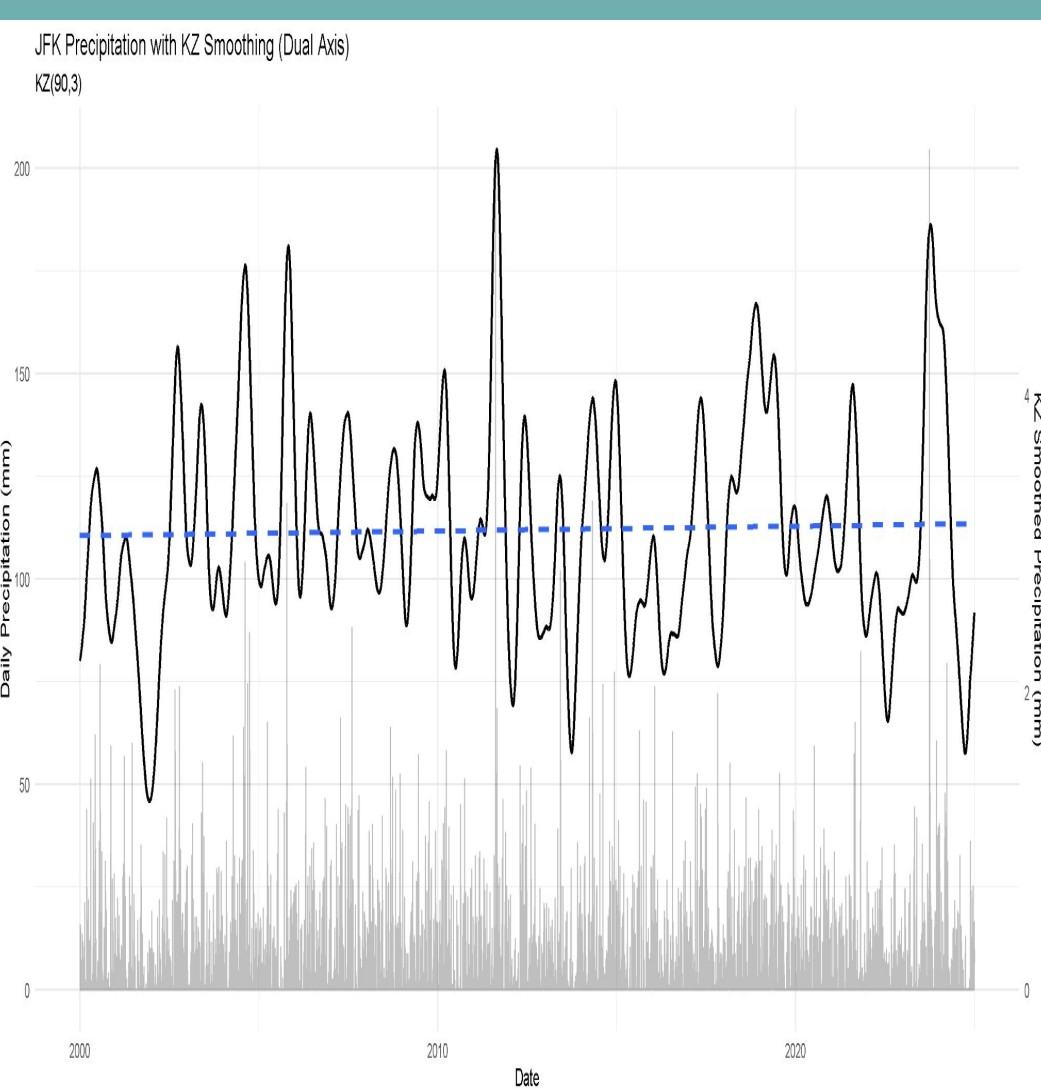


Figure 9: Daily precipitation at JFK (2000–2024) shown as gray bars, with a Kolmogorov–Zurbenko (KZ) smoothed series ($m = 90$, $k = 3$) overlaid as a black line (right axis).

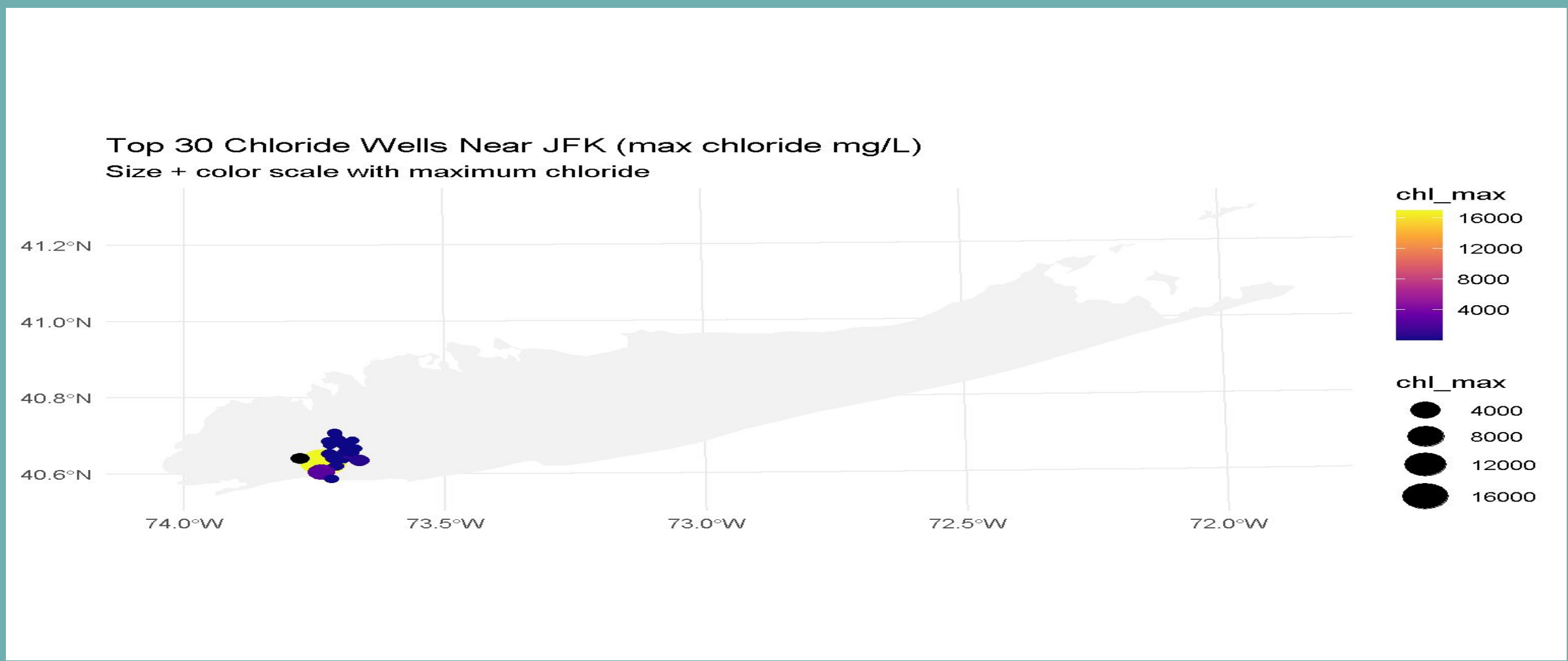


Figure 2: Spatial distribution of the 30 groundwater wells with the highest recorded chloride concentrations near JFK Airport (within 10 km radius). Point size and color represent maximum observed chloride concentration (mg/L), with larger and warmer-colored symbols indicating higher values. Wells are clustered within the highly urbanized coastal zone, highlighting areas of elevated salinity and guiding selection of the representative groundwater time series used for spectral analysis.

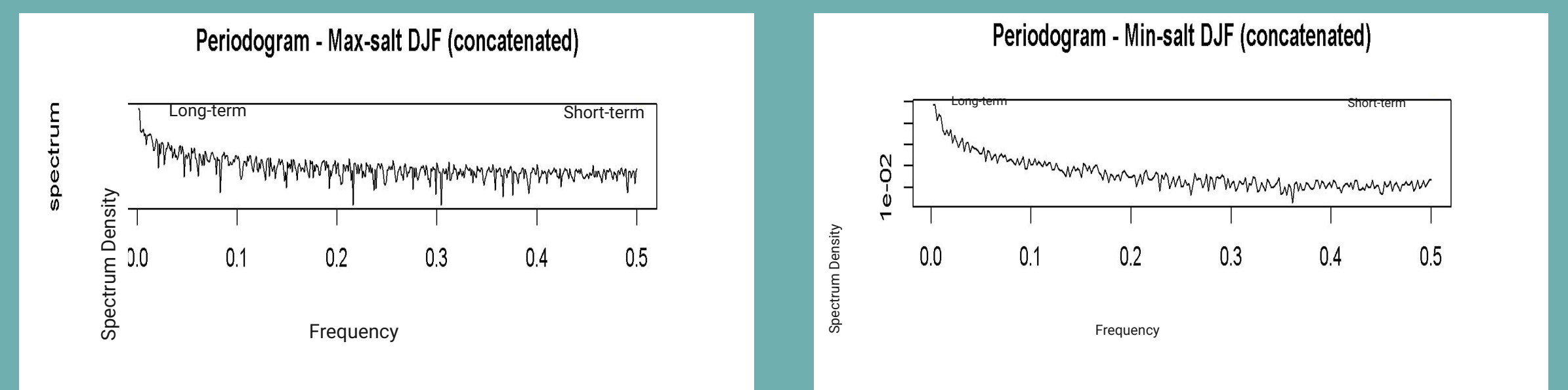


Figure 3: Periodogram analysis of groundwater levels during DJF periods showing differences in frequency structure between minimum-salt and maximum-salt winters.

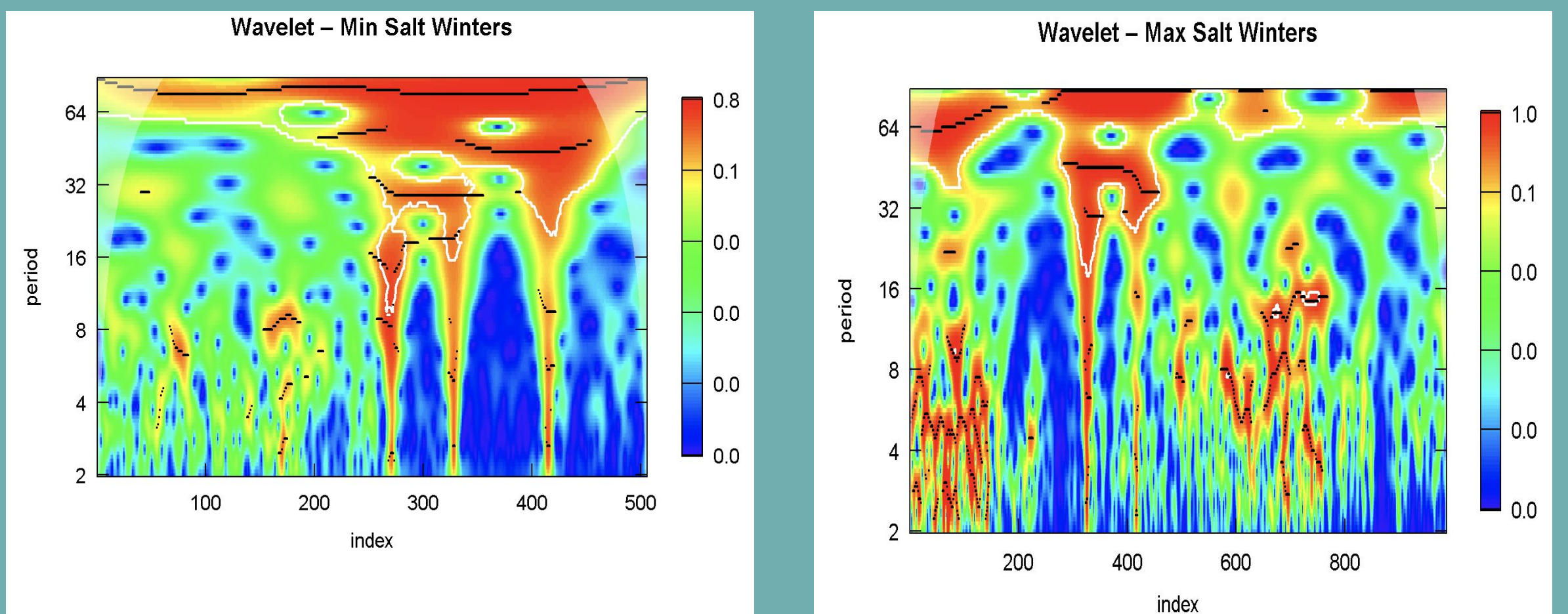


Figure 4: Wavelet power spectrum of groundwater levels during minimum-salt winters, showing dominant spectral energy at longer periods (~ 30 – 90 days) and a continuous band indicative of stable, seasonally driven variability.

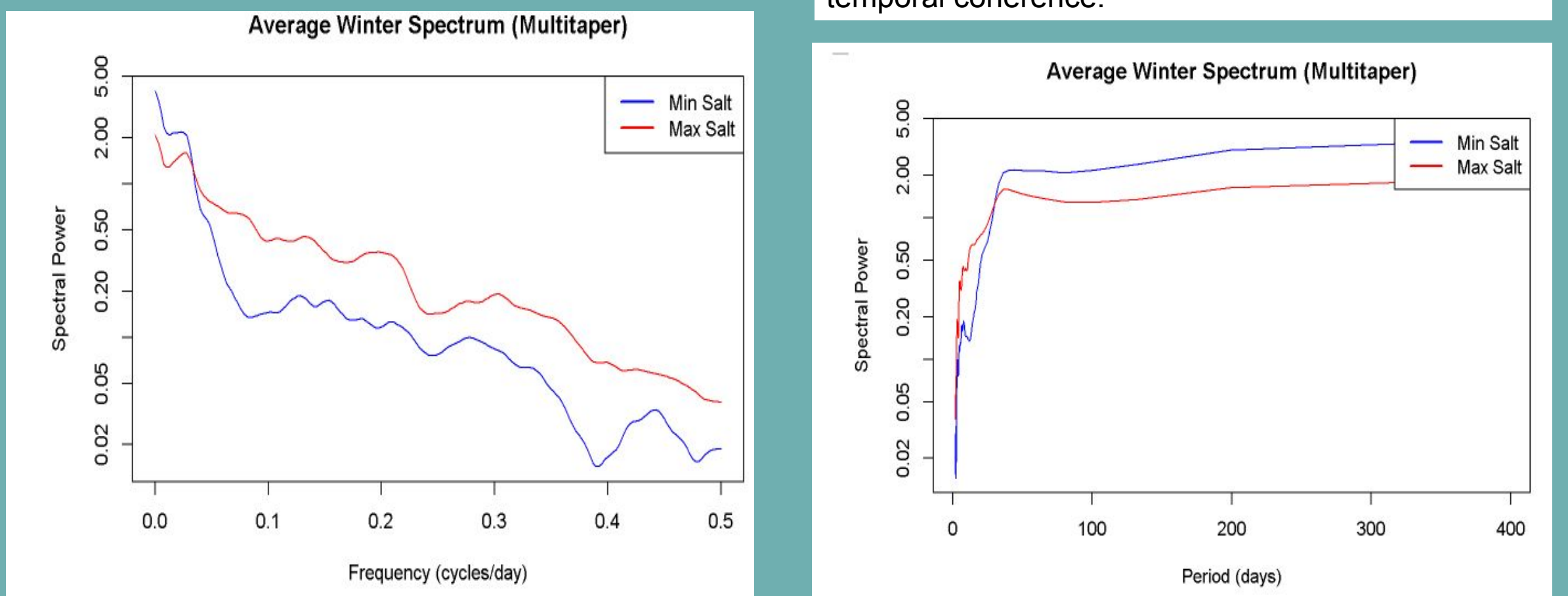


Figure 6: Frequency-domain spectral comparison of groundwater levels during minimum-salt and maximum-salt winters, showing reduced low-frequency power and a flatter spectral distribution under maximum-salt conditions.

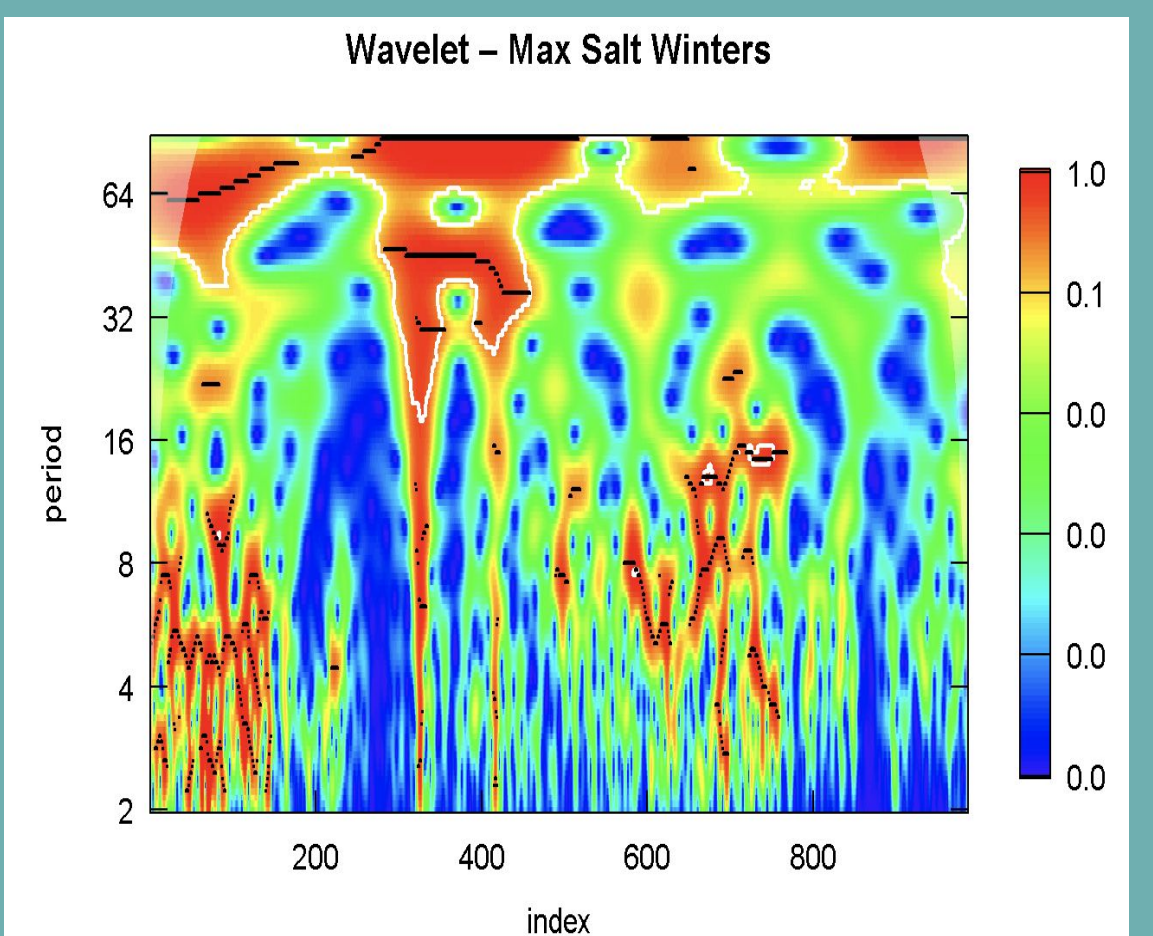


Figure 5: Wavelet power spectrum of groundwater levels during maximum-salt winters, showing a broader and more fragmented distribution of spectral power across both long and short periods, indicating increased short-term variability and reduced temporal coherence.

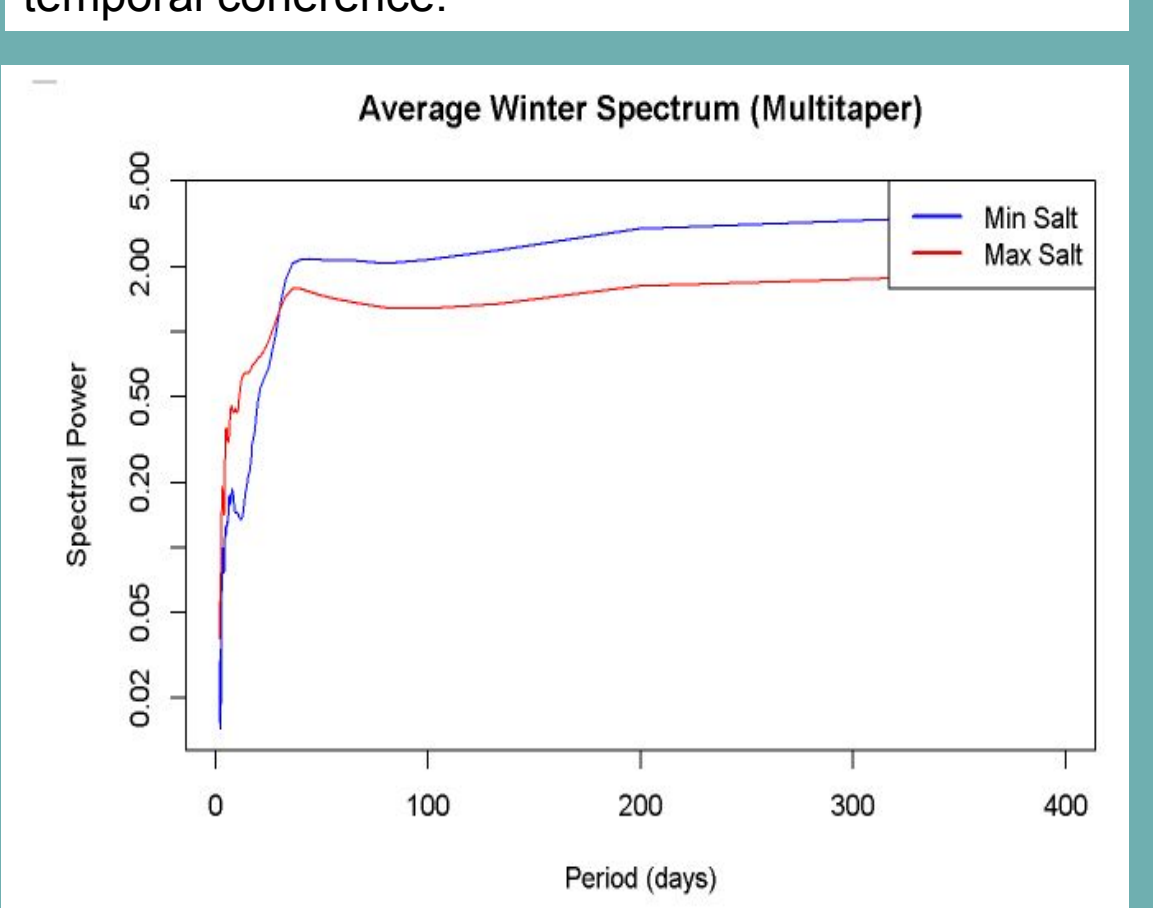


Figure 7: Period-domain spectral comparison of groundwater levels, illustrating reduced spectral power at longer periods (~ 100 – 300 days) during maximum-salt winters, indicating diminished seasonal variability.

Discussion/Conclusion:

- Groundwater near JFK varies with winter conditions: frequent freeze–thaw $\rightarrow$ higher short-term variability; fewer freeze events $\rightarrow$ more stable, buffered response
- Winter processes (road salt, freeze–thaw, urban runoff) influence groundwater dynamics, though temperature was used as a proxy (not direct salt data)
- Spectral analysis shows a shift from long-term, storage-dominated behavior (low-salt) to short-term, event-driven responses (high-salt), with a transition at ~ 30 – 40 days
- Precipitation shows no long-term trend, suggesting it is not the primary driver of observed groundwater variability
- Overall, winter conditions significantly alter groundwater behavior, highlighting the need to include seasonal processes in management and future studies with direct salt data.

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Credit Authorship Contribution Statement:

Neville, J.: Editing, writing- Results, Conclusion, Probatory R-studio Coding-figure 7, unused CCF lag graphs, research groundwork, R-Studio reviewing; Marsellos, A.E.: supervision, Rstudio coding, guidance, editing, Discussion; Tsakiri, K.G.: Rstudio coding review; Mighty, M.: peer researcher, research groundwork, writing- discussion, editing; Bernhardt, L.: writing - abstract and introduction, research, Rstudio coding, revising.

References:

- De Cicco, L. A., Hirsch, R. M., Lorenz, D., & Watkins, W. (2023). *dataRetrieval: R packages for discovering and retrieving water data available from U.S. federal hydrologic web services*. U.S. Geological Survey.
- Sparks, A. H., & Hengl, T. (2023). *GHCN: Download and process NOAA Global Historical Climatology Network data*. R package. National Weather Service. *Climate*. NOAA. <https://www.weather.gov/wrh/Climate?wfo=okx>
- Parag, A., Marsellos, A. E., Tsakiri, K. G., et al. (2025). *Wavelet analysis of groundwater level fluctuations in Long Island*. Long Island Geology Conference.
- Torrence, C., & Compo, G. P. (1998). *A practical guide to wavelet analysis*. Bulletin of the American Meteorological Society, 79(1), 61–78.